

DATE:

EXPT NO:04

DATA VISUALIZATION

AIM:

To visualize the Student Performance dataset using Python and create different charts to understand student academic performance and identify important patterns.

PROCEDURE:

1. Start the program.
2. Import **Pandas, Matplotlib, and Seaborn** libraries.
3. Load the Student Performance CSV dataset.
4. Display the first few records using head().
5. Create a **bar chart** to compare students based on performance.
6. Create a **pie chart** to show the distribution of students by gender.
7. Create a **line chart** to analyze the relationship between study hours and performance.
8. Add suitable titles, axis labels, and legends.
9. Display all the visualizations.
10. Analyze the graphs and identify important patterns.
11. Stop the program.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("student_performance_data.csv")

# Select numerical columns
num_cols = df.select_dtypes(include="number").columns
```

```

print("Numerical columns:", num_cols.tolist())

# ----- BAR CHART -----
plt.figure(figsize=(8,5))

df[num_cols[0]].value_counts().head(10).plot(kind="bar")

plt.title("Bar Chart")
plt.xlabel(num_cols[0])
plt.ylabel("Count")
plt.show()

# ----- BOX PLOT -----
plt.figure(figsize=(8,5))

sns.boxplot(x=df[num_cols[0]])

plt.title("Box Plot")
plt.xlabel(num_cols[0])
plt.show()

# ----- LINE CHART -----
plt.figure(figsize=(8,5))

plt.plot(df[num_cols[0]].head(20),
         marker="o")

plt.title("Line Chart")
plt.xlabel("Student")
plt.ylabel(num_cols[0])
plt.grid()
plt.show()

```

OUTPUT:

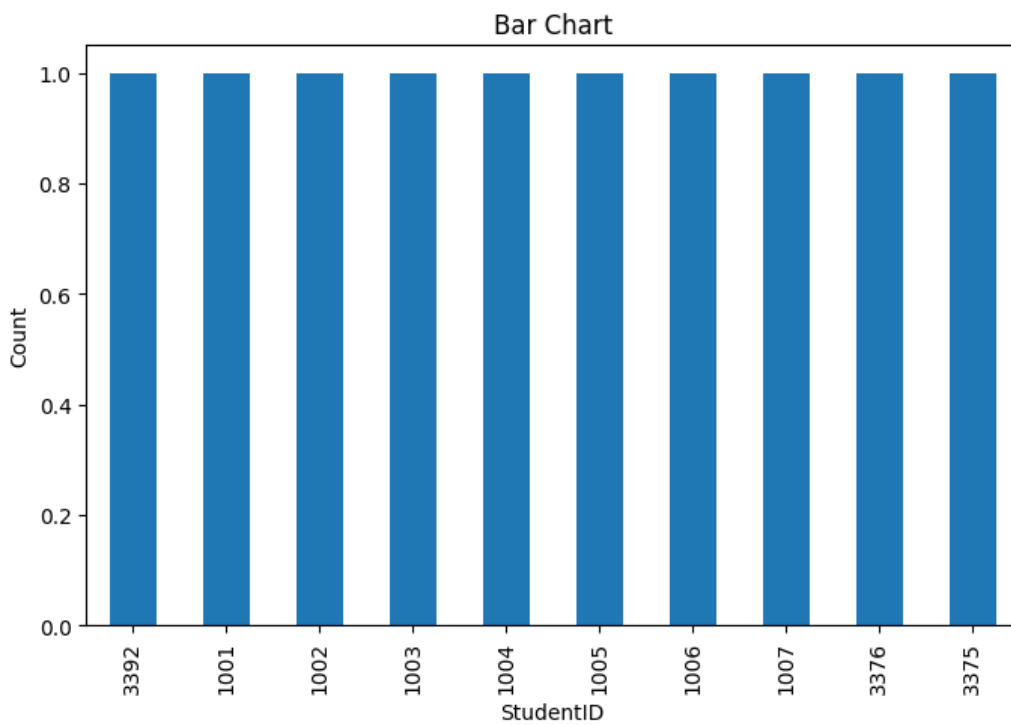
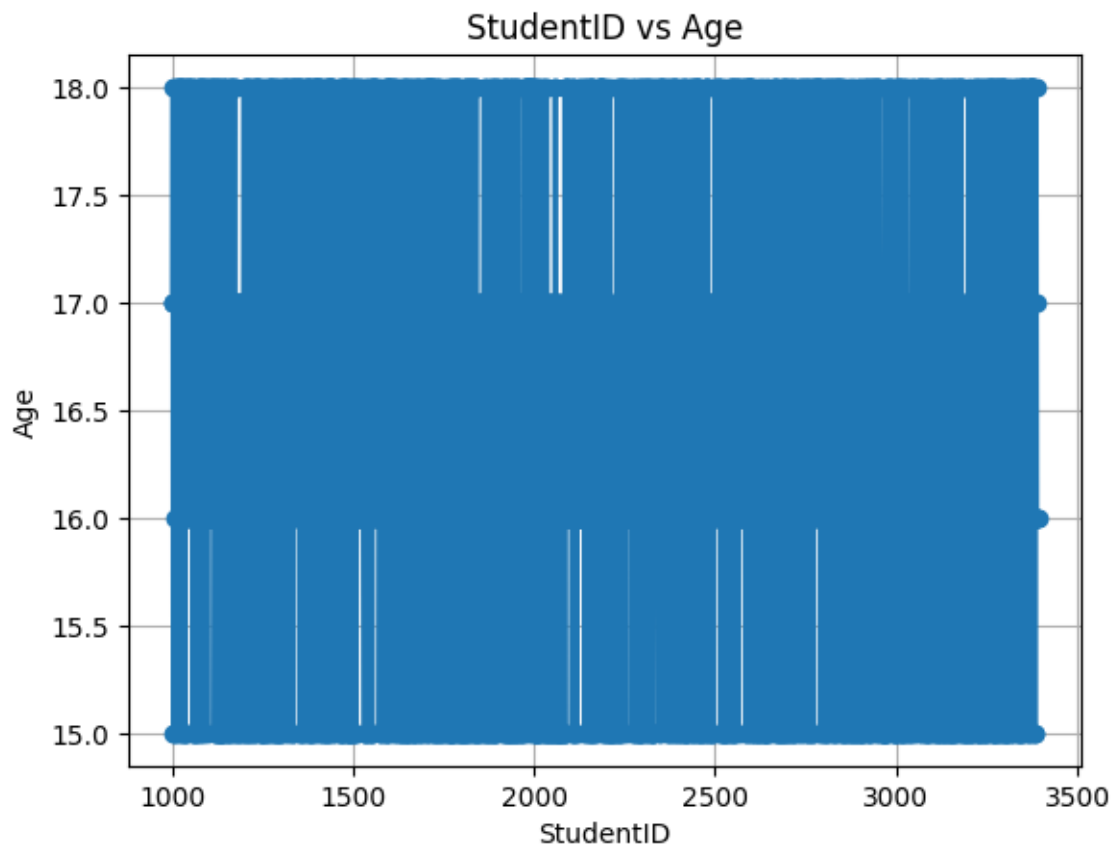
Columns:

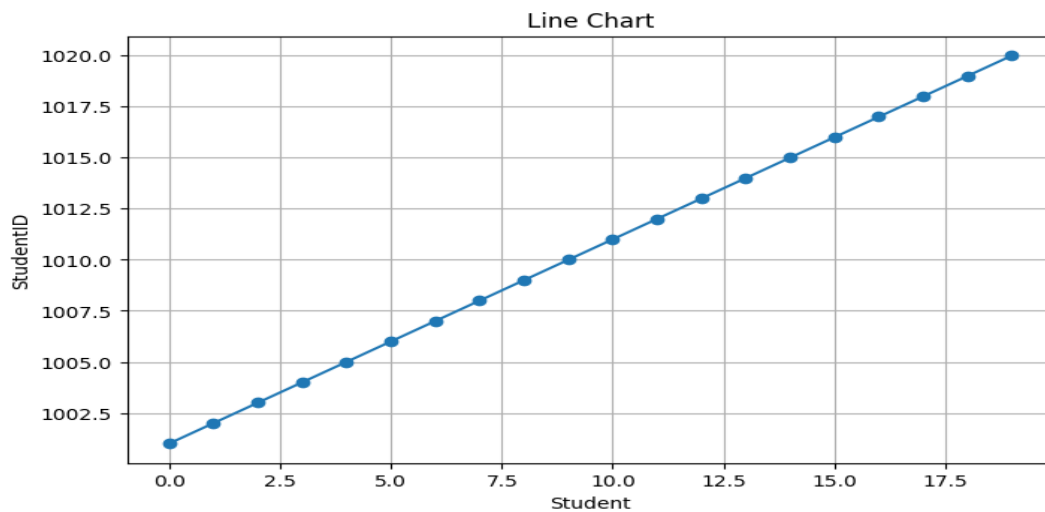
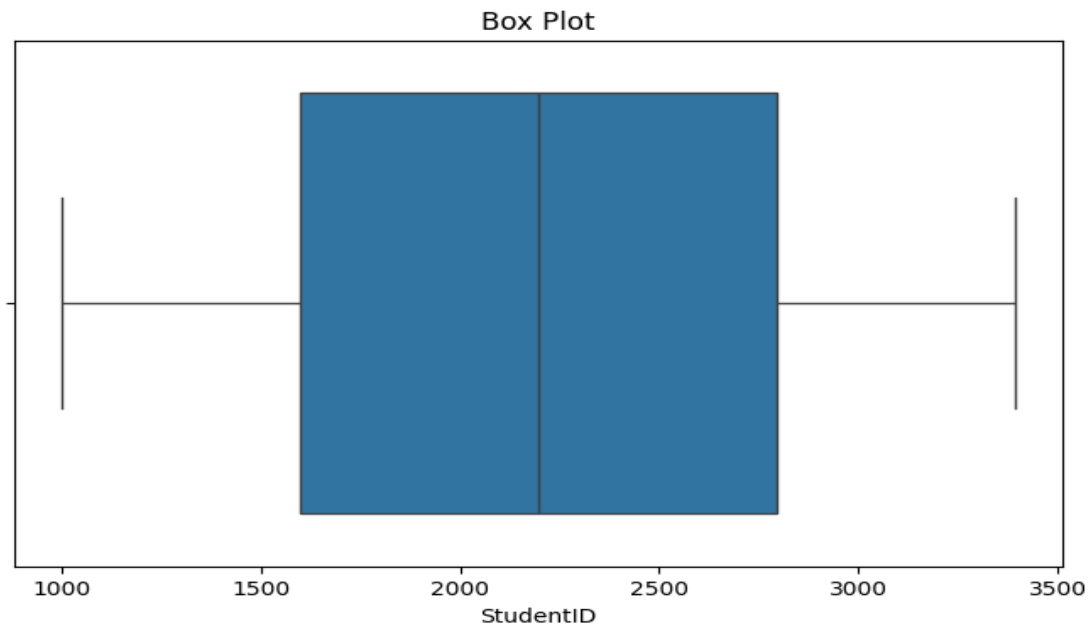
['StudentID', 'Age', 'Gender', 'Ethnicity', 'ParentalEducation',
 'StudyTimeWeekly', 'Absences', 'Tutoring', 'ParentalSupport',
 'Extracurricular', 'Sports', 'Music', 'Volunteering', 'GPA', 'GradeClass']
 StudentID Age Gender Ethnicity ParentalEducation StudyTimeWeekly

\						
0	1001	17	1	0	2	19.833723
1	1002	18	0	0	1	15.408756
2	1003	15	0	2	3	4.210570
3	1004	17	1	0	3	10.028829
4	1005	17	1	0	2	4.672495

	Absences	Tutoring	ParentalSupport	Extracurricular	Sports	Music	\
0	7	1	2	0	0	1	
1	0	0	1	0	0	0	
2	26	0	2	0	0	0	
3	14	0	3	1	0	0	
4	17	1	3	0	0	0	

	Volunteering	GPA	GradeClass
0	0	2.929196	2
1	0	3.042915	1
2	0	0.112602	4
3	0	2.054218	3
4	0	1.288061	4





RESULT:

The customer churn dataset was successfully imported, analyzed, visualized, and processed. A Logistic Regression model was trained to predict customer churn, and the model performance was evaluated using accuracy and a confusion matrix. The processed data was also exported to an Excel file.